PRIMER · FOR RESIDENTS

Community Primer

A plain-language starting point. What a data center is, why companies choose a community, who builds them, and what they do to a place, plus a glossary and resources.

From the Grade My Data Center research team

What is a data center?

Data centers are the physical buildings where anything digital lives. Every time you watch a video online, search the Internet, or use an AI tool, your phone or computer has to retrieve information. That information is stored in powerful computers called servers that are stacked on shelves in a data center.

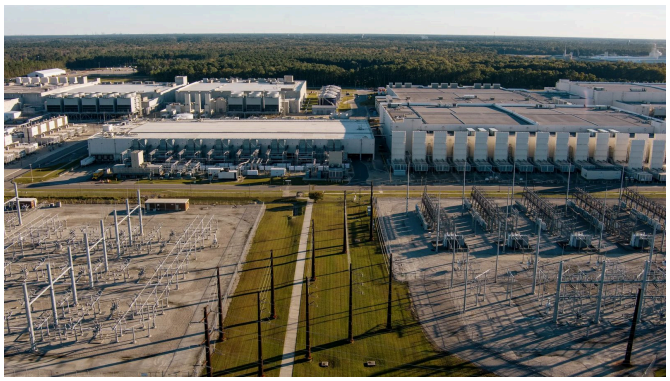
Why would companies want to put a data center in my community?

Companies primarily put data centers in places that have the resources needed to run the data center, are close to customers, and have local policies that welcome data center developments. Companies point to three main resources they need for data centers: available power, access to water, and proximity to fiber / Internet connections.

What does a data center look like?

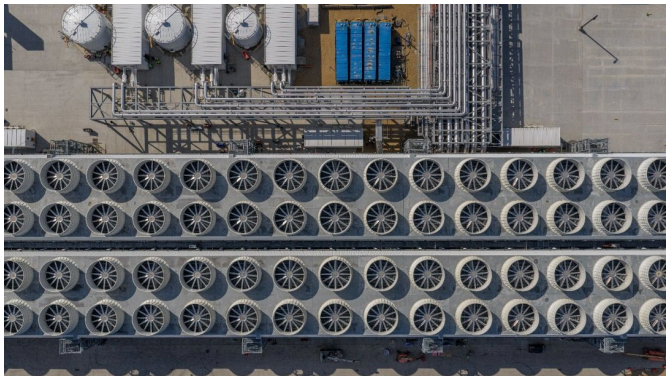
You might not even know when you have walked or driven by a data center. They usually look like windowless boxes. They often have security fences, security cameras, and lights that stay on all night. Data centers also have power equipment that makes a steady humming noise. They might have a few parking spots for employees, but they generally look like a cross between a grocery store and a shipping warehouse - just without the people and trucks.

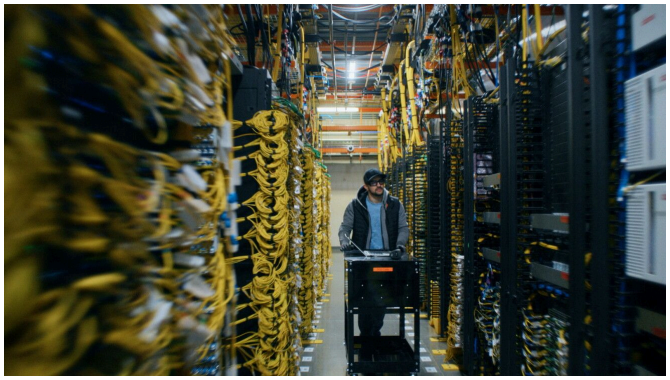










Images sourced from the corporate websites of Google, Amazon, Meta, and Microsoft.

Hyperscale data centers are super-sized data centers. They hold anywhere from 5,000 to hundreds of thousands of servers. These massive campuses usually have several buildings: an office building for a small staff, multiple buildings to hold the servers, and equipment to handle power and water. These giant campuses are mostly used by big technology companies you know, like Amazon, Google, Microsoft, or Meta.

Hyperscale data centers take up a lot of space. That's why they're often built in places with a lot of open land. In those areas, a campus can spread across hundreds or thousands of acres (an acre is a little bigger than a football field).

Very large hyperscale data centers can be mind-bogglingly huge. For example, one hyperscale data center in Louisiana is 2,500 acres, five-miles long and almost one-mile wide.[1] That is more than twice the size of the closest airport, 20 times the size of the Mall of America in Minnesota, and one-tenth the size of Disney World in Orlando, FL. If you built homes on that land instead of data centers, you could build more than 2,500 standalone houses or 50,000 apartments.

When data centers are built in suburbs or cities, they have to fit into tighter spaces. Because of this, they might be several stories tall and sit right next to the street. Sometimes, companies buy old industrial warehouses or factories and convert them to data centers.

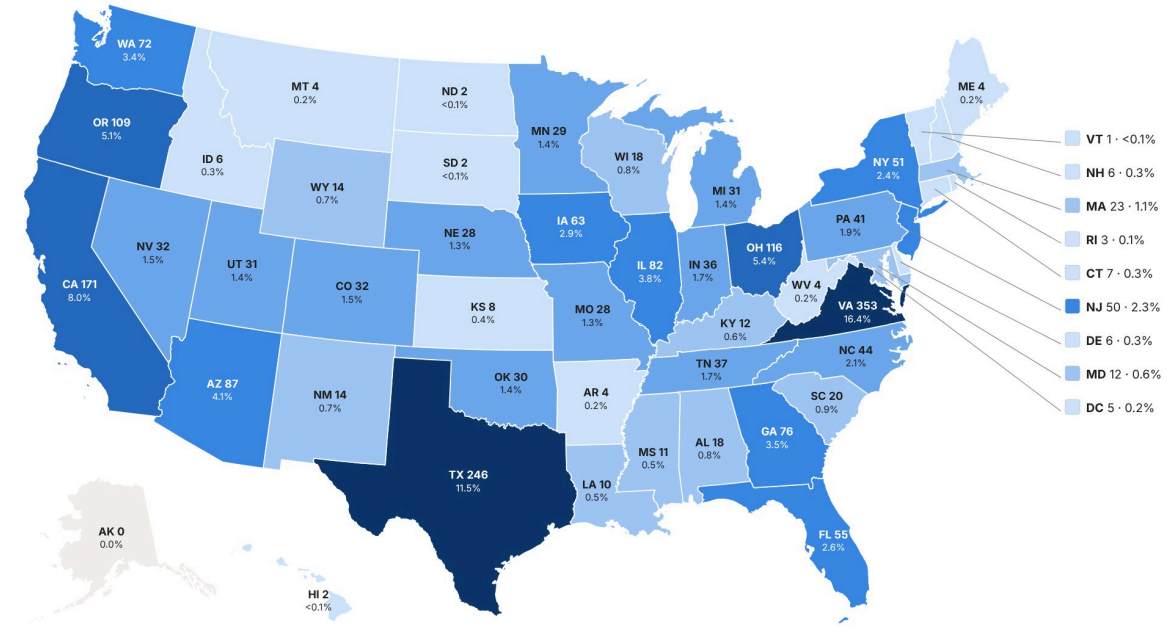
Where are data centers?

No one knows the exact number of data centers in the U.S., but estimates range between 2,000 and 5,400 facilities. There is no single list. However, we do know there are data centers in every single state except Alaska. Virginia, Texas, and California have the most. Virginia alone has more than 15% of all U.S. data centers.

There are another 4,400 data centers proposed to be built soon. Many will go to Virginia, Texas, and California. Hundreds more are planned for states like Georgia, Pennsylvania, Arizona, Illinois, and Ohio.

Active data center facilities by state

2,146 active U.S. facilities · label shows count and share of national total



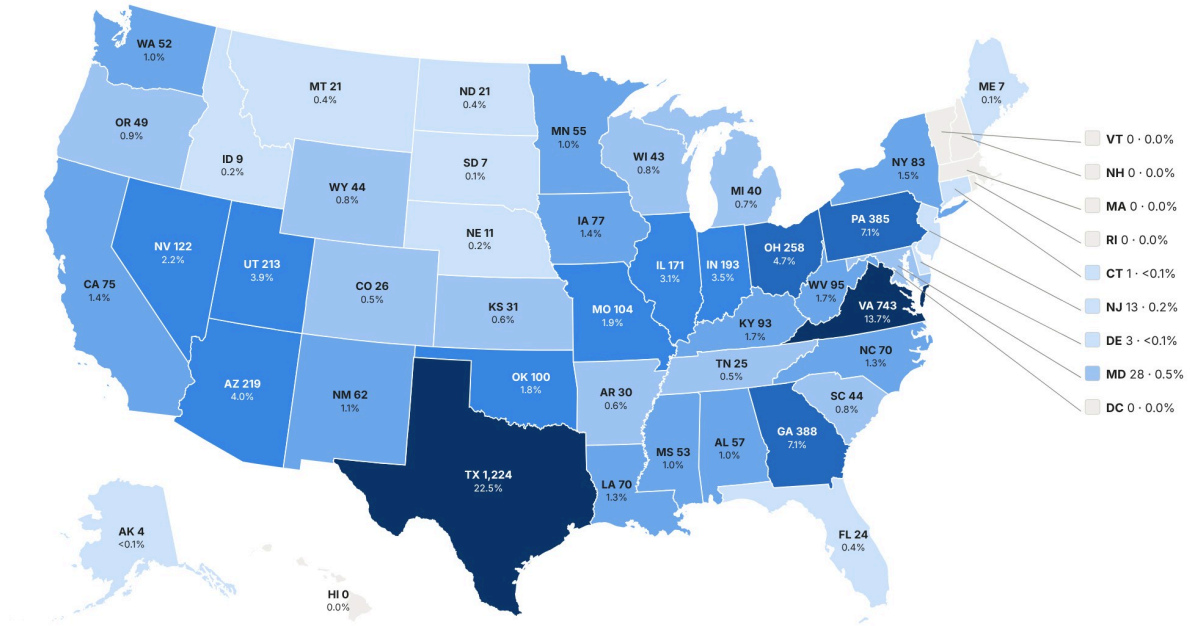
Active facilities

0 | 1–9 | 10–24 | 25–49 | 50–99 | 100–199 | 200+

Source: Aterio, Accessed 4 November 2026. Facilities are individual buildings. Campuses may have more than one building, so this map may overcount data center campuses.

Under construction, delayed, and announced data center facilities by state

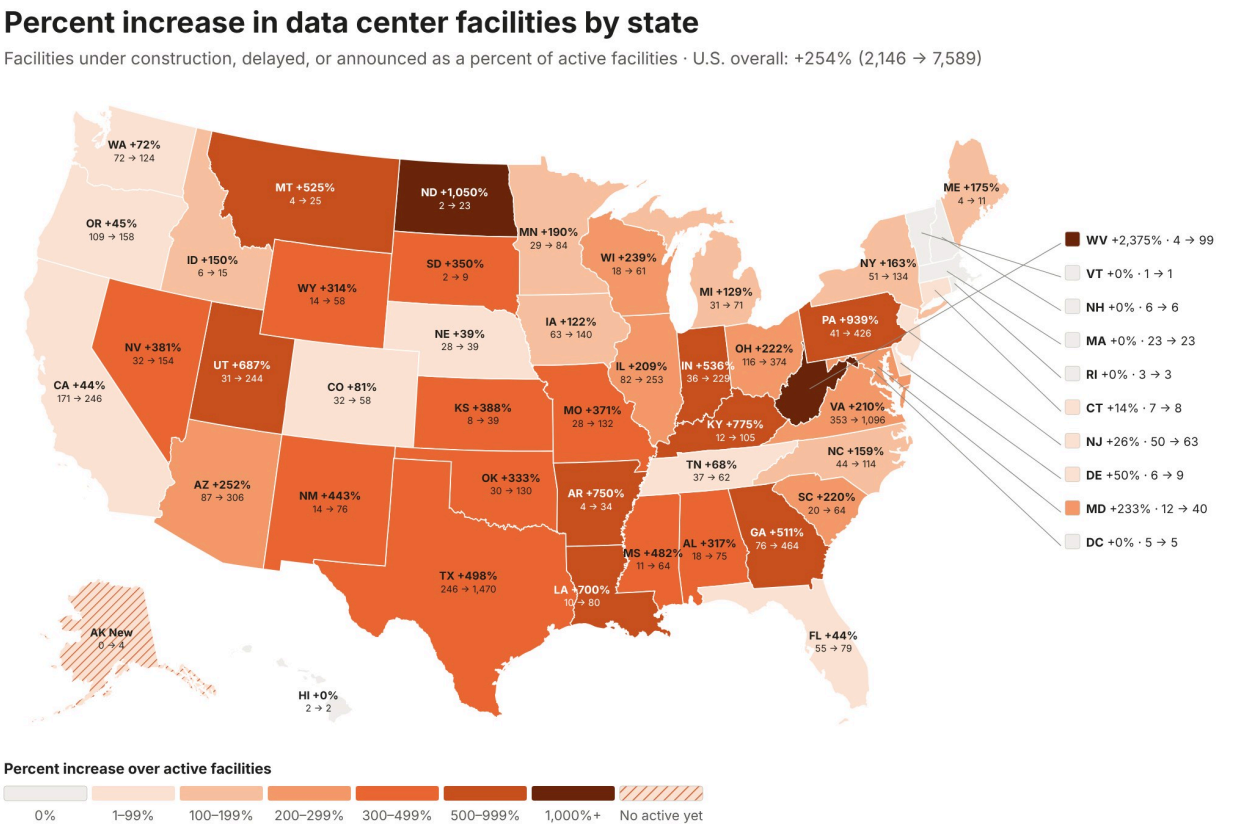
5,443 U.S. facilities: under construction (890), delayed (88), or announced (4,465) · label shows count and share of national total



Facilities under construction, delayed, or announced

0 | 1–24 | 25–49 | 50–99 | 100–249 | 250–499 | 500+

Source: Aterio, Accessed 4 November 2026. Facilities are individual buildings. Campuses may have more than one building, so this map may overcount data center campuses.

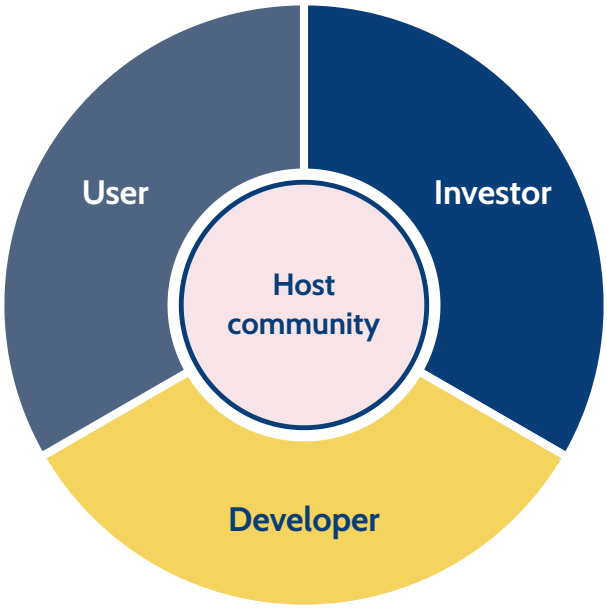


Labels show percent increase and active facilities → active + under construction, delayed, and announced. Alaska has no active facilities (0 → 4).
Source: Aterio, Accessed 4 November 2026. Facilities are individual buildings. Campuses may have more than one building, so this map may overcount data center campuses.

Who builds data centers?

Every data center project needs four main things to exist: 1) an investor who pays for the project, 2) a developer who constructs the project, 3) a user who owns, rents, or runs the servers inside, and 4) a host community, the actual place where the data center is built. Sometimes, one company plays the first three roles all by themselves. For example, tech giants like Amazon or Google will often buy the land, build the buildings, and run their own computers inside. Other times, three or more different companies have to team up as partners. One company might pay for it, another might build it, and a third company might rent the building to use for their computers. When companies team up like this, they often create a "shell company" that has a different name they use on local government permits and applications.

EVERY DATA CENTER PROJECT NEEDS



Investor
The entity who pays for the project.

Developer
The entity who constructs the project.

User
The entity who owns, rents, or runs the servers inside.

Host community
The place where the data center is built and the people that "host" it.

What are the impacts of data centers?

Energy & electricity

The servers in data centers consume a lot of energy and give off a lot of heat. A single hyperscale data center can use more than 100 megawatts (MW) of power. That is the same amount of energy to power more than 100,000 homes or charge up to 3,000 electric cars at the same time. This power usually comes from the local grid, the same electric utility that gives your home power. Some data centers generate their own power on-site.

Heat

When thousands of servers run at the same time, they give off heat. Most of this heat is cooled down by large air conditioners inside the building before it is released outside. However, research shows that letting this warm air out can still raise the local outdoor temperature, creating a warm pocket of air called a heat island.

Water usage

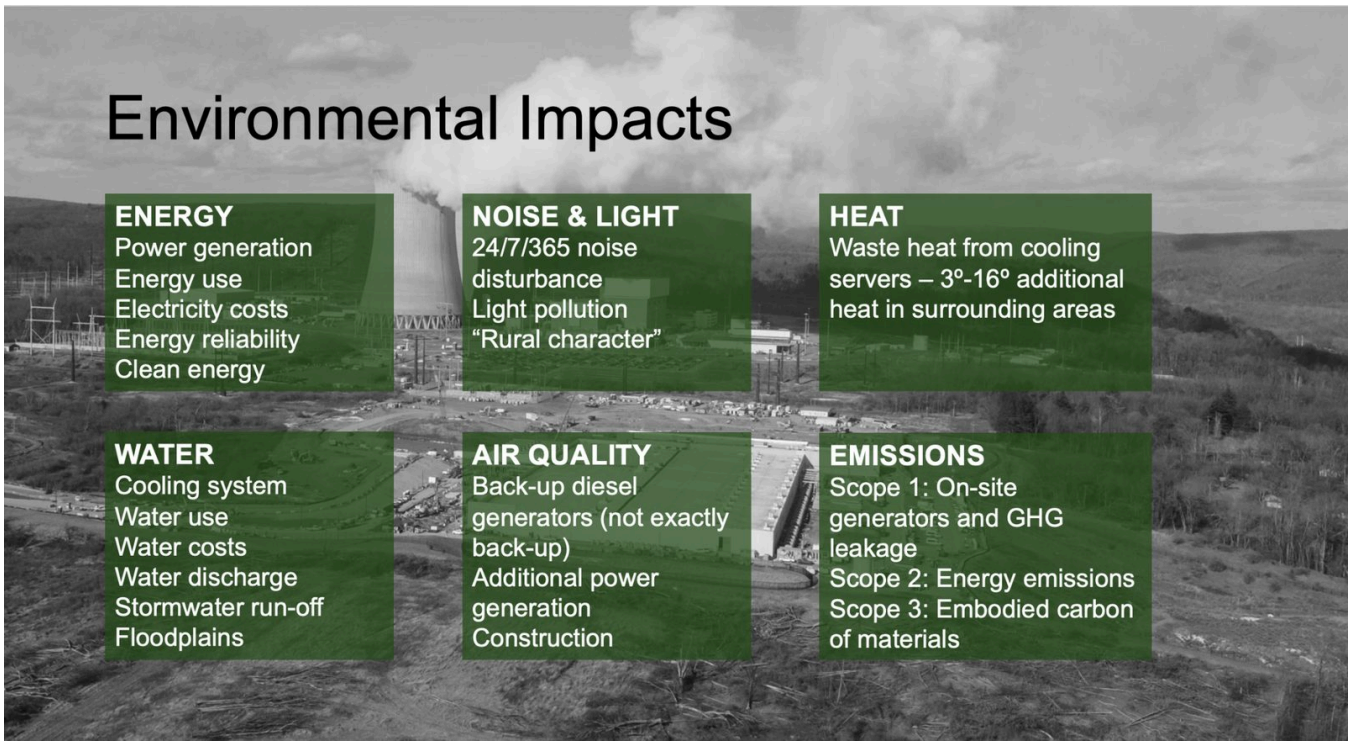
Data centers can use water in two ways. First, they use it to cool down hot servers. Depending on the cooling system they use, a data center might use millions of gallons of water per day (like filling multiple swimming pools) or they might use a closed-loop system that recycles water so it only uses as much as a toilet flush. Second, a lot of water is used at the power plants that make the facility's electricity. This water generally comes directly from the same local water utilities that supply your home.

Noise & light

Data centers never sleep. They run 24 hours a day, 365 days a year. Because of this, their giant cooling fans and electrical equipment make a constant, low humming noise. Neighbors say it sounds like a refrigerator or hairdryer that never turns off. To keep the buildings safe, bright security lights stay on all night.

Jobs

Data centers bring different kinds of jobs. During construction, they create hundreds or thousands of temporary jobs for builders. Because hyperscale campuses are so big, construction can last for several years. Once the building is finished, however, it is mostly run by computers. A finished hyperscale data center only needs a couple hundred permanent workers to keep it running. Smaller data centers need even fewer workers.



Environmental impacts



Social impacts



Design impacts

Glossary

Campus

A group of buildings that all belong to the same data center. Think of it like a school campus, where several buildings sit on the same piece of land and work together for the same purpose.

Closed-loop system

A way of cooling computers that reuses the same water over and over again, kind of like a fish tank that recirculates its water. This uses much less water than systems that drain the water away after one use.

Data center

A large building full of powerful computers that store digital information. When you watch a video, send a text, search online, or query AI, the information you need comes from a data center somewhere.

Developer

A company that builds data centers or other buildings. Sometimes the same company builds the data center and uses it. Other times, one company builds it and a different company moves in.

Electric grid

The system of power lines and equipment that carries electricity from power plants to homes, schools, and businesses. It works kind of like a network of roads, but for electricity instead of cars.

Electric utility

The company that supplies electricity to homes and businesses in your area.

Energy storage

A way of saving electricity to use later, usually in very large batteries.

Fossil fuels

Fuels like coal, oil, and natural gas that come from the ground and are burned to make electricity. They are called "fossil" fuels because they formed underground over millions of years. Some data centers burn fossil fuels to make their own electricity.

Heat island

A small area that gets warmer than the land around it because of heat coming from buildings or machines. Data centers can create heat islands because they constantly blow hot air out of their cooling systems.

Host community

The neighborhood, town, or city where a data center is built.

Hyperscale data center

A very large data center that can hold thousands or even hundreds of thousands of servers. These are usually owned and used by big tech companies like Amazon, Google, Microsoft, or Meta.

Investor

A person or company that pays for a project to be built, hoping to make money from it later.

Light pollution

Too much artificial light at night that makes the sky brighter than it should be. It can bother people who live nearby and disrupt animals that depend on darkness.

Megawatt (MW)

A unit used to measure electricity. One megawatt can power about 1,000 average homes. Some large data centers use more than 100 megawatts, enough electricity for more than 100,000 homes.

Renewable energy

Energy that comes from natural sources that never run out, like sunlight, wind, and water.

Server

A powerful computer that stores information and sends it to other devices when they need it. Servers are what make the internet work. Data centers are basically giant warehouses full of servers, stacked on shelves in long rows.

Shell company

A business that a larger company creates and gives a different name to, often used when applying for permits or buying land.

Glossary built with the assistance of Google Gemini Pro.

Resources

WATER

Data centers require massive amounts of water to keep servers from overheating. This report reveals that two-thirds of new data centers are popping up in areas already facing severe water shortages, putting local reservoirs at risk. Additionally, ordinary residents often end up footing the bill on their monthly utility rates. Learn more at the link below and see how communities are responding.

Water & Data Centers

ELECTRICITY

This policy brief by the Union of Concerned Scientists found that utilities passed more than $4.3 billion in direct connection costs onto households across seven states in 2024 alone. When data centers connect to the grid we all use, utilities sometimes pass infrastructure costs on to everyone's monthly rates instead of charging the tech companies. Learn more about how data centers may impact electricity bills at the link below.

Union of Concerned Scientists

LEARN FROM OTHER COMMUNITIES

This policy toolkit lays out concrete legal tools, such as building moratoriums, strict zoning setbacks, noise limits, and bans on corporate non-disclosure agreements, that could be helpful for communities at the local, state, regional, and federal levels. There are also examples of policies ranked either very strong, strong, or weak. View the toolkit at the link below.

North Star Data Center Policy Toolkit

WORKING WITH LOCAL ELECTED OFFICIALS

This discussion guide is designed for communities to figure out whether a project is truly beneficial or merely extractive. It also lays out clear action steps for tracking local zoning proposals, working with elected leaders, and securing enforceable community benefits before ground is broken. If you're looking for a framework to start conversations about data centers, check out the link below.

Data Center Discussion Guide

FROM THE HORSE'S MOUTH: DEVELOPER WEBSITES

There are a number of major developers of hyperscale data centers in the US. View links to their data center pages and environmental reports below.

Microsoft	**Data centers**	**Sustainability info**
Google	**Data centers**	**Sustainability info**
Meta	**Data centers**	**Sustainability info**
AWS	**Data centers**	**Sustainability info**

Notes

Meta Facebook page for Richland Parish data center, accessed 20 July 2026, **facebook.com/RichlandParishDataCenter**